



OSCAR PRATOLA

Aerospace Engineer, M.Sc. — Aeronautics, Space & Defense

✉ oscar.pratola.eng@gmail.com | oscar-pratola | Portfolio

Nationality: Italian | Como, Italy | Available to relocate

Driving license: B, A2, Boat License | **UAV/Drone Pilot License** (EU OPEN A1/A3) |

Profile

Aeronautical engineer (M.Sc., Politecnico di Milano) with hands-on R&D experience spanning **aeronautics, space, and defense-relevant systems**: propulsion, energetic materials, additive manufacturing, and structural design. Delivered a novel thermite-loaded polymer manufacturing route for spacecraft Design-for-Demise within the EU-funded THREAD consortium (**Horizon-EIC-2024-PATHFINDER**), from formulation to combustion testing, alongside project work on liquid rocket engines, missile-class vehicle design, turbofan propulsion, and ESA-supervised flight hardware. **ESA Academy alumnus** and **M.B.D.A. Defense Leadership Academy** graduate, with a structured, safety-conscious approach to hazardous materials and test-campaign risk management. **Author of 3 conference papers**. Seeking a **hands-on engineering** role across aerospace or defense in a **dynamic, innovative** and **multicultural** environment.

Highlights:

Aeronautical Engineering M.Sc. | EU Horizon Projects | ESA Academy Alumnus | Defense Leadership Academy Alumnus | Aerospace Propulsion | Energetic Materials | Additive Manufacturing | Design-for-Demise | CAD/FEM/CFD | Hazardous Materials Handling | Test Engineer | Hands-On Skills | Team Managing | System Engineering | V-Model Verification | Root Cause Analysis

Experience

⚙️ **R&D Engineer — Space Propulsion Laboratory (SPLab), Politecnico di Milano** 10.2024 – 10.2025

EU-funded THREAD project (Horizon-EIC-2024-PATHFINDER) — Design-for-Demise energetic materials

- Led experimental research on thermite-loaded photocurable resins for spacecraft demise, from slurry formulation to DLP 3D-printing process optimization (**technology investigation** → **process optimization** → **characterization**)
- Developed a repeatable **Advanced Manufacturing methodology** via Vat Photopolymerization (**Additive Manufacturing**) of **energetic composite materials** to enable satellite demise
- Designed development and testing workflow via **V-Model** approach and performed **Root-Cause Analysis** to address project criticalities
- Designed chemical passivation treatments for reactive metallic powders, preventing premature crosslinking of the polymer matrix
- Performed multi-technique material characterization (**TG/DTA, FTIR, SEM/EDX, Laser Diffraction, Computed Tomography**) and ignition/combustion testing (**Pyrometry, High-Speed Imaging**)
- Managed material procurement and safety risk assessments; interfaced with international partners (**Airbus Defence & Space, DLR, CEIIA, ENEA**); co-authored **3 conference publications**

⚙️ **Technical Advisor — Additively Manufactured Micro-Thrusters** 06.2024 – 09.2024

MEC-3D (Internship)

- Conducted feasibility study on metal AM micro-thrusters for SmallSat/CubeSat platforms, benchmarking Hall-Effect Thrusters vs. resistojets (Isp, thrust-to-power, TRL)
- Proposed a dual-mode Additively Manufactured Vaporizing Liquid Microthruster (green propellant + cold-gas backup) and screened candidate superalloys (Inconel 718/625) for SLM
- Delivered a technical feasibility report and R&D roadmap to industrial stakeholder leadership

⚙️ **Team Leader — PoliSpace Dissemination Department** 04.2024 – 03.2025

Departmental science communication team (10 members)

- Coordinated a multidisciplinary team of writers, designers, and social media specialists; managed workflows and public relations for the departmental science blog

Education

📖 **M.Sc. Aeronautical Engineering — Politecnico di Milano** 09.2022 – 10.2025

Experimental Thesis: Shaping Thermites: Investigation of Casting and 3D Printing for Spacecraft Demise Applications

📖 **B.Sc. Aerospace Engineering — Politecnico di Milano** 09.2017 – 09.2022

Core disciplines: fluid mechanics, structural analysis, propulsion, materials science

📖 **Diploma, Chemistry, Materials & Biotechnologies — I.T.I.S. Paolo Carcano** 09.2012 – 07.2017

Analytical, organic and industrial chemistry; extensive laboratory practice

Technical Skills

- **Propulsion, Aerodynamics & Energetics:** Space & aeronautical propulsion, electric propulsion, aircraft systems, energetic materials, combustion testing & chemistry, thermochemistry (CEA), risk assessment.
- **Additive Manufacturing & Materials:** FDM, SLA/DLP, VAT polymerization, metal AM (PBF/DED), composite materials, DfAM, topology optimization, manufacturing process optimization.
- **Experimental & Characterization:** TG/DTA, FTIR, SEM/EDX, laser diffraction, pyrometry, high-speed imaging, computed tomography.
- **Engineering Analysis & Systems:** CFD, FEM, fracture mechanics, root cause analysis, V-Model verification & validation, reporting & technical documentation.
- **Safety & Compliance:** Hazardous materials handling, risk assessment, regulatory/range-safety screening, test-campaign planning.
- **CAD & Modeling Software:** SolidWorks, Autodesk Inventor, Autodesk Fusion, AutoCAD, Altair Inspire, nTop, Abaqus, Ansys Granta, OpenFOAM.
- **Coding & Software Utilities:** MATLAB (proficient), Simulink, Python, C, $\LaTeX$, 3D printing slicers, SpectraGryph, NETZSCH Proteus.
- **Hands-on & Workshop:** Workshop machining and tooling, electrical troubleshooting, woodwork.

Selected Projects

Preliminary Design of an ICBM for Rapid Payload Delivery (Course Project, MSc) — Supervised by Prof. F. Maggi; conceptual scaling and trajectory optimization of a sub-orbital 3-stage solid rocket motor launcher (Orion 50XL / STAR 37FM / STAR 30BP) for a 100 kg cargo payload, utilizing Modified Newtonian Flow Theory and ANSYS Fluent CFD codes to map supersonic/hypersonic aerodynamics at max-q.

Preliminary Design of a Semi-Cryogenic Liquid Rocket Engine (Course Project, MSc) — LOX/RP-1 blowdown liquid motor (1 kN): discretized MATLAB model of transient fluid-dynamics, discharge coefficient sensitivity under metal Additive Manufacturing (LPBF) surface roughness uncertainties, and regenerative cooling jacket analysis.

Topology Optimization of a Vega Payload Adapter Bracket (Course Project, MSc) — Supervised by **Dr. T. Ghidini** (ESA Head of Materials & Processes); ECSS-compliant structural redesign and optimal printing orientation simulation (Altair Inspire/Netfabb) of a monolithic AlSi10Mg component.

A-10 Thunderbolt II Propulsion Analysis (BSc Thesis) — Supervised by Prof. C. Paravan; numerical MATLAB modeling suite simulating the baseline Joule-Brayton cycle of the TF34-GE-100 turbofan engine, validating compressor aerodynamic stall boundaries.

Static and Fatigue Analysis of a Helicopter Rotor Mast (Course Project, MSc) — Supervised by Prof. M. Giglio; analytical and numerical (Abaqus FEM) structural assessment of a main rotor shaft, correlated with experimental strain gauge rosette data via MATLAB.

Certifications & Training

- **E.S.A. Academy** — Space Debris Training Course (2025): fragmentation, collision risk, mitigation guidelines, design-for-demise
- **M.B.D.A. Defense Leadership Academy** — School of the National Interest: conflict zones, battle theaters, and defense technologies, with engineers from Leonardo S.p.A., Rheinmetall AG, and MBDA
- **UAV/Drone Pilot License** — EU OPEN Category A1/A3
- **Space Systems & Advanced Materials:** Additive Manufacturing Specialization — ASU/Coursera; Space Architecture Workshop — School of Disruption; Materials Science for Advanced Technological Applications — ASU/Coursera; Introduction to Molecular Spectroscopy — Univ. of Manchester/Coursera
- **Systems Engineering:** Systems Engineering — MathWorks/Coursera
- **Aerospace Engineering:** Sustainable Aviation: The Route to Climate-Neutral Aviation — DelftX; Introduction to the Flight Mechanics of Aerobatics — Aero Club Milano
- **Programming & Data Analysis:** Crash Course on Python — Google/Coursera; Data Analysis with Python — Univ. of Michigan/Coursera

Publications

- F. Maggi, A. Finazzi, J. Domaschio, **O. Pratola**, et al., "Current Developments in Polymer-Based Reactive Materials for Thermite-for-Demise (T4D) Technology," *AIAA SciTech 2026 Forum*, 2026.
- A. Finazzi, **O. Pratola**, C. Zanardi, et al., "Thermite-for-Demise (T4D): Thermite Charge Shaping Advancements," *11th EUCASS*, 2025.
- O. Pratola**, S. Carlotti, J. Domaschio, et al., "Vat Polymerization Additive Manufacturing of Thermite Charges for Satellite Demise Applications," *14-ISICP*, 2025.

Languages

Italian (Native) | **English** (Fluent) | **French** (Basic)